

Q1 Solution:

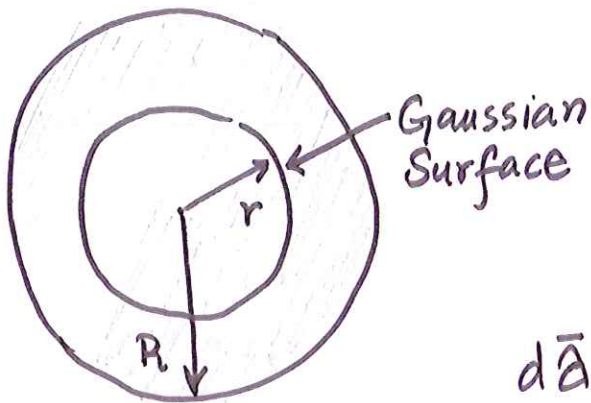
An easier approach is to use Gauss's Law:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$$

Since the density is uniform, therefore

$$Q_{enc} = \text{Volume} \times \text{Volumetric density}$$

$$Q_{enc} = \frac{4}{3}\pi r^3 \rho$$



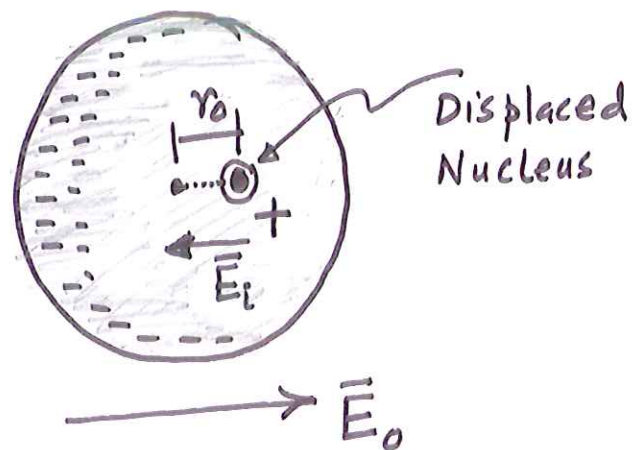
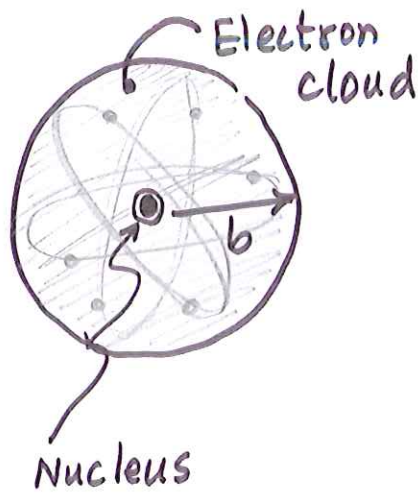
$$d\vec{a} = \text{differential area} = r^2 \sin\theta d\theta d\phi \hat{r}$$

$$\vec{E} = E_r \hat{r}$$

$$\oint_S \vec{E} \cdot d\vec{a} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} r^2 E_r \sin\theta d\theta d\phi = 4\pi r^2 E_r$$

$$\Rightarrow E_r = \frac{\rho r}{3\epsilon_0} \Rightarrow \vec{E} = \frac{\rho r}{3\epsilon_0} \hat{r}.$$

Q2. Solution.



Force exerted by external field on the nucleus is $\vec{F}_0 = N|e|\vec{E}_0$

Force exerted by electron cloud on the nucleus is $\vec{F}_i = N|e|\vec{E}_i$

- $\vec{E}_i$ is the electric field experienced at r_0 by the electron cloud

From Q1: $E = \frac{\rho r_0}{3\epsilon_0}$ where $\rho = \frac{N|e|}{\frac{4}{3}\pi b^3}$

At equilibrium, the magnitude of the two electric field ($\vec{E}_0$ and $\vec{E}_i$) will be equal.

$$E_0 = \frac{\rho r_0}{3\epsilon_0} \quad \text{where } \rho = \frac{N|e|}{\frac{4}{3}\pi b^3}$$

$$Q3. \quad \vec{F} = xy \hat{x} - 2x \hat{y}$$

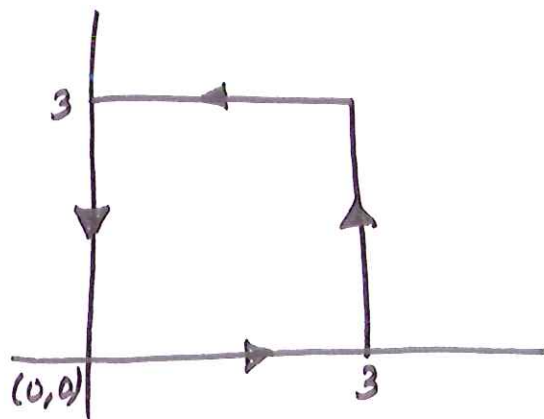
$$\nabla \times \vec{F} = (-2-x) \hat{z}$$

$$\int_S (\nabla \times \vec{F}) \cdot d\vec{a} =: LHS.$$

$$d\vec{a} = dx dy \hat{z}$$

$$\int_S (\nabla \times \vec{F}) \cdot d\vec{a} = \int_0^3 \int_0^3 (-2-x) dx dy = -63/2.$$

$$\oint \vec{F} \cdot d\vec{l} = 0 + (-18) + (-27/2) + 0 = -63/2.$$



Q4.

$$V = - \int \vec{F} \cdot d\vec{l}$$

$$= - \int (3y dx + (3x - 2z) dy - (2y + z) dz)$$

$$= - (3xy + 3xy - 2zy - 2yz - z^2/2) + C$$

$$= - 6xy + 4yz + z^2/2 + C$$

Q5.

The component of $\vec{A}$ in the direction of $\vec{B}$

$$\text{is } \left(\frac{\vec{A} \cdot \vec{B}}{\vec{B} \cdot \vec{B}} \right) \vec{B}$$

The component of $\vec{B}$ in the direction of $\vec{A}$

$$\text{is } \left(\frac{\vec{A} \cdot \vec{B}}{\vec{A} \cdot \vec{A}} \right) \vec{A}$$

Look @ the solution of HW#1
at LMS for detail work.

Q6 Solution. I.

since $x = 2y^2$
therefore $dx = 4y dy$

$$\int_{P_1}^{P_2} \vec{E} \cdot d\vec{l} = \int_{(2,1)}^{(8,2)} (y dx + x dy)$$

$$= \int_{(2,1)}^{(8,2)} (4y^2 dy + 2y^2 dy)$$

$$= \int_1^2 6y^2 dy = \left. \frac{6y^3}{3} \right|_1^2 = 2(2^3 - 1^3) = 14.$$

Q7. Solution.

$$\vec{r}' = z \hat{z} \quad (z \text{ is changing from } -L/2 \text{ to } +L/2)$$

$$\vec{r} = h \hat{z}$$

$$dq = \lambda dz$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{|\vec{r} - \vec{r}'|}$$

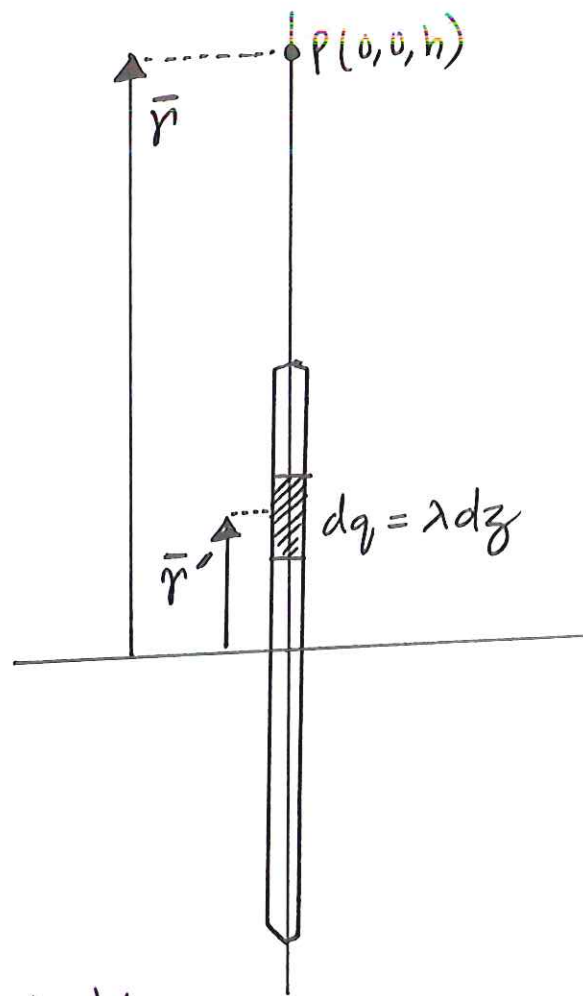
$$V(h) = \frac{1}{4\pi\epsilon_0} \int_{-L/2}^{+L/2} \frac{\lambda dz}{h - z}$$

$$\text{Let } h - z = u \Rightarrow -dz = du$$

$$V(h) = \frac{-\lambda}{4\pi\epsilon_0} \int_{h+L/2}^{h-L/2} \frac{du}{u} = \frac{\lambda}{4\pi\epsilon_0} \int_{h-L/2}^{h+L/2} \frac{du}{u}$$

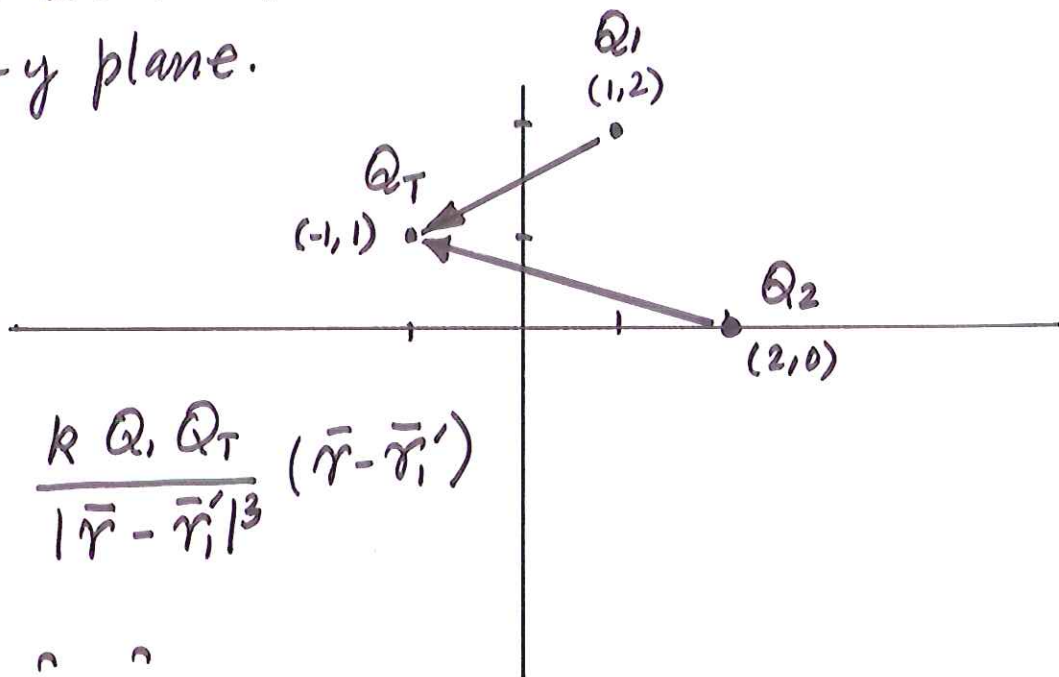
$$= \frac{\lambda}{4\pi\epsilon_0} \ln u \Big|_{h-L/2}^{h+L/2}$$

$$V(h) = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(h+L/2) - \ln(h-L/2) \right]$$



Q8. Solution.

Note that the z -component is zero for all three locations, so we can realize the positions on x - y plane.



$$\vec{F}_1 = \frac{k Q_1 Q_T}{|\vec{r} - \vec{r}_1'|^3} (\vec{r} - \vec{r}_1')$$

$$\vec{r} = -\hat{x} + \hat{y}$$

$$\vec{r}_1' = \hat{x} + 2\hat{y}$$

$$\vec{r}_2' = 2\hat{x} + 0\hat{y}$$

$$\vec{F}_1 = \frac{k Q_1 Q_T (-\hat{x} + \hat{y} - \hat{x} - 2\hat{y})}{(\sqrt{(-2)^2 + (-1)^2})^3}$$

$$\vec{F}_1 = \frac{k Q_1 Q_T (-2\hat{x} - \hat{y})}{(5)^{3/2}}$$

$$\vec{F}_2 = \frac{k Q_2 Q_T (-\hat{x} + \hat{y} - 2\hat{x})}{(\sqrt{(-3)^2 + (1)^2})^3}$$

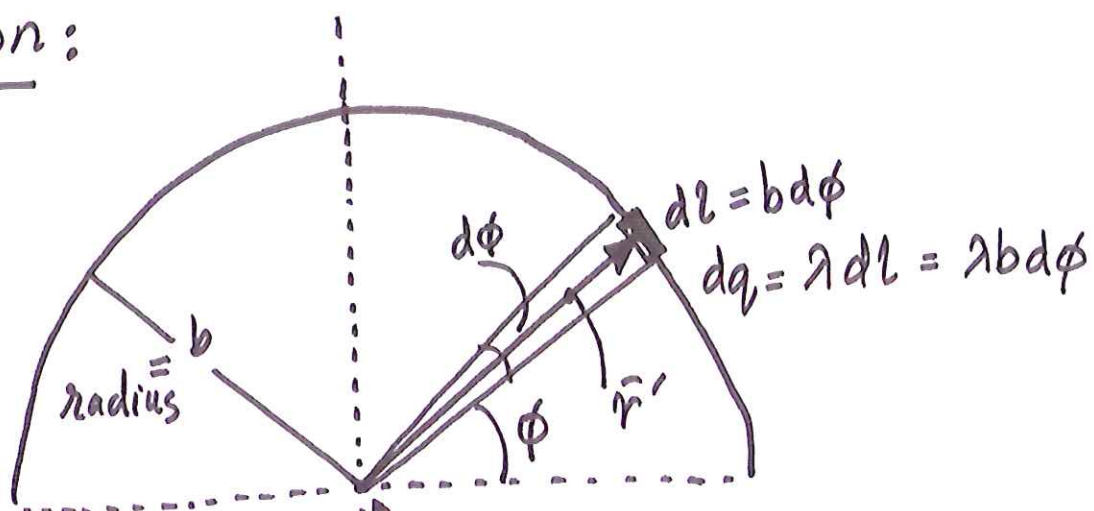
$$\vec{F}_2 = \frac{k Q_2 Q_T (-3\hat{x} + \hat{y})}{(10)^{3/2}}$$

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = \frac{k Q_T}{(5)^{3/2}} \left(Q_1 (-2\hat{x} - \hat{y}) + Q_2 \frac{(-3\hat{x} + \hat{y})}{(2)^{3/2}} \right)$$

(a) when x -component is zero: $2Q_1 = -\frac{3Q_2}{2\sqrt{2}} \Rightarrow \boxed{Q_1 = -\frac{3}{4\sqrt{2}} Q_2}$

(b) when y -component is zero: $\boxed{Q_1 = -\frac{Q_2}{2\sqrt{2}}}$

Q9. Solution:



$$\vec{r}' = b \hat{r} = x \hat{x} + y \hat{y}$$

$$\begin{aligned} x &= b \cos \phi \\ y &= b \sin \phi \end{aligned}$$

Field point is at the origin
thus, $\vec{r} = 0 \Rightarrow \vec{E} = -\vec{r}'$

Note that owing to symmetry
x-component of the electric field
experienced at the origin will be
cancelled out.

Only the y-component will survive, and
they all add up when the differential
charge dq is integrated from $\phi = 0$ to
 $\phi = 180^\circ = \pi$ radians.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$= \frac{1}{4\pi\epsilon_0} \int_{\phi=0}^{\phi=\pi} \frac{\lambda b d\phi (-x\hat{x} - y\hat{y})}{b^3}$$

Note:

$$dq = \lambda dl$$

$$dq = \lambda b d\phi$$

$$\vec{r} = 0$$

$$\vec{r}' = x\hat{x} + y\hat{y}$$

$$\vec{r}' = b\hat{\rho}$$

$$|\vec{r} - \vec{r}'| = |\vec{r}'| = b$$

Since, x-component will be cancelled out, we only need to keep y-component and drop x-component. Substitute $y = b \sin\phi$.

$$\vec{E} = -\frac{1}{4\pi\epsilon_0} \int_0^\pi \frac{\lambda \sin\phi d\phi \hat{y}}{b}$$

$$= \frac{\lambda}{4\pi\epsilon_0 b} \cos\phi \Big|_0^\pi \hat{y} = \frac{-\lambda \hat{y}}{2\pi\epsilon_0 b}$$

$\vec{E}$ has only y-component which is in downward direction along y-axis.